

Daily Progress Report — Day 4

Project Name: MediFlow CRM — Doctor / Clinic Practice Management System

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Developer/Reporter: Debarati Gantait, Anwesha Manna, Indranil Dutta

Reporting Period: Daily

1. Summary of Today's Activities

Today, the team focused on the complete end-to-end development and verification of **Phase 5: Appointments Module & Interactive Calendar**, strictly aligned with **Section 4 (Pages 3–4)** and **Section 12 (Pages 14–15)** of the Clinic CRM System.pdf specification. We engineered the appointment database schema, built the automated APT-XXXXX identifier generator, and implemented full lifecycle status transitions (Pending → Confirmed → Checked In → In Consultation → Completed / Cancelled / No Show). We designed and delivered 5 responsive Blade templates including an interactive **Month, Week, and Day Calendar view**, and integrated live appointment history into the 9-Section Patient Profile Hub. In addition, we finalized the architectural foundation and database migrations for **Phase 6: Clinical Consultations & Digital Prescriptions**. All work was styled with Laravel Pint, assets re-bundled with Vite, and verified via an expanded automated test suite of **42 tests passing with 0 failures**.

2. Key Discussions & Decisions

A. Appointments Scheduling & Lifecycle Engine (Phase 5)

- **Database Architecture & Model Hooks:** Created and executed migration for appointments table (2026_09_17_070711_create_appointments_table.php) with foreign keys to patients and users (doctors), scheduling timestamps, visit reasons, preliminary clinical notes, 7 lifecycle states, 3 payment statuses, cancellation audit reasons, and soft deletion. Built Appointment.php with automated sequential APT-XXXXX code generation upon record creation.
- **Clinic Constraints Integration:** Integrated practice slot duration (15 min default), opening/closing hours (09:00 to 19:00), and daily appointment quotas dynamically loaded from the Setting repository.
- **Lifecycle Workflow Actions:** Built single-click state transition endpoints enabling front-desk and doctors to advance visits from arrival (Check In) to clinical encounter (Start Consultation) and closure (Complete / Cancel / No-Show).

B. Frontend & Interactive Calendar Development (Phase 5)

- **Interactive Calendar Navigation:** Developed appointments/calendar.blade.php featuring three synchronized views:
 - **Month View:** 7x5 responsive grid displaying scheduled appointments with color-coded status chips, day counters, and quick booking links.

- **Week View:** 7-column time grid with scheduled visit cards and daily add-slot triggers.
- **Day View:** Hourly timeline schedule from clinic opening to closing hours with direct patient check-in actions.
- **Top KPI Cockpit & Filter Engine:** Created appointments/index.blade.php with 6 live metric cards for today's visits, instant toggle between List and Calendar views, multi-criteria search, and date presets (Today, Tomorrow, This Week, Upcoming, Past, Custom).
- **Patient Profile Hub Integration:** Integrated live appointment history into **Tab 3 (Appointment History)** of patients/show.blade.php, displaying chronological patient visits, status pills, and direct scheduling shortcuts.

C. Phase 6 Architectural Foundation (Consultations & Prescriptions)

- **Clinical Encounters & Prescription Schema:** Designed and created database migrations for:
 - consultations table (2026_09_17_082138_create_consultations_table.php): Stores presenting symptoms, provisional diagnosis, structured vitals JSON (BP, Pulse, Temperature, SpO2, Weight, Height, BMI), examination findings, and treatment recommendations.
 - prescriptions table (2026_09_17_082139_create_prescriptions_table.php): Unique RX-XXXXX numbering, clinical notes, diagnostic tests ordered, dietary/lifestyle advice, and follow-up date.
 - prescription_items table (2026_09_17_082140_create_prescription_items_table.php): Itemized medication dosing (name, dosage, frequency, duration, route, timing, and instructions).

D. Quality Assurance & Code Standards

- Authored tests/Feature/AppointmentTest.php with 12 comprehensive feature tests verifying CRUD, role permissions, auto-ID generation, lifecycle transitions, calendar sub-views, and patient profile tab rendering.
- Full test suite execution: **42 passed, 0 failed, 137 assertions** (100% pass rate).
- Formatted all modified PHP files using Laravel Pint (vendor/bin/pint --format agent).
- Rebuilt production frontend assets with Vite (npm run build).

3. Members Role

Team Member	Assigned Role	Module & Responsibility Breakdown
Anwesh Manna <i>(Backend & Database)</i>	Backend Architecture & Database Lead	• Created and executed database migration for appointments and Phase 6 migrations (consultations, prescriptions, prescription_items) • Implemented Appointment.php Eloquent model with

		automated APT-XXXXXX generator and bidirectional relationships (Patient, User) • Developed AppointmentController.php with multi-criteria search, date presets, and status transition pipeline • Created AppointmentSeeder.php populating realistic multi-status visit schedules • Authored PHPUnit automated feature tests for the Appointments module
Indranil Dutta (<i>Full-Stack / Clinical Workflow Lead</i>)	Clinical Engine & Workflow Specialist	• Designed and implemented the appointment booking form (appointments/create.blade.php) and edit modal (appointments/edit.blade.php) with dynamic slot generation • Engineered the Appointment Lifecycle Cockpit (appointments/show.blade.php) linking patient allergy alerts with visit progress actions • Integrated real-time appointment records into Tab 3 of the 9-Section Patient Profile Hub (patients/show.blade.php) • Designed the clinical encounter schema and vitals capture structure for Phase 6 Consultations
Debarati Gantait (<i>Frontend, Billing & Reporting Lead</i>)	Billing, Analytics & UI/UX Lead	• Built the interactive calendar interface (appointments/calendar.blade.php) supporting Month, Week, and Day timeline views • Styled the Appointments index table with responsive desktop table and mobile card stack transformations • Designed the top KPI metric cards and quick action transition badges • Executed Vite frontend asset compilation (npm run build) and Laravel Pint code styling audits

4. Roadblocks & Knowledge Gathering Strategy

- **Area of Focus:** Designing an interactive, multi-view calendar (Month / Week / Day) that seamlessly supports practitioner filtering and fast date navigation without requiring bulky external calendar libraries.
- **Action Taken:** Leveraged Carbon date arithmetic on the backend paired with lightweight Alpine.js reactivity and Tailwind grid utility classes, achieving zero additional external JS dependencies while maintaining sub-second render speeds.
- **Next Focus:** Completing **Phase 6 (Clinical Consultations & Digital Prescriptions)** by implementing ConsultationController, PrescriptionController, the dynamic Alpine.js medicine repeater, printable Rx template, and patient hub Tabs 4 & 5.

Submitted By GROUP A

(Anwesha Manna, Debarati Gantait, Indranil Dutta)